Doolittle Decomposition

From Compact LU Factorization to Partial Pivoting

Pengju Jin Siquan Fang Haoye He

Zhejiang University

October 6, 2026

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Why LU Factorization?

Solve $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{R}^{n\times n}$ nonsingular, for many $\mathbf{b}$:

$$A = LU \implies \underbrace{L\mathbf{y} = \mathbf{b}}_{\text{forward}}, \quad \underbrace{U\mathbf{x} = \mathbf{y}}_{\text{back}}.$$

Decompose *once*; each new $\mathbf{b}$ then costs only $O(n^2)$.

Roadmap

Basic formulas & compact storage $\rightarrow$ shortcomings $\rightarrow$ partial pivoting ($PA = LU$) $\rightarrow$ correctness.

The Doolittle Convention

LU is not unique: $LU = (LD)(D^{-1}U)$ for any nonsingular diagonal D. Fix the scale by making L *unit* lower triangular ($\ell_{ii} = 1$):

$$L = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ \ell_{21} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ \ell_{n1} & \cdots & \ell_{n,n-1} & 1 \end{bmatrix}, \qquad U = \begin{bmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & u_{nn} \end{bmatrix}.$$

For nonsingular A (when the factorization exists), this makes L, U **unique**.

Recursive Formulas

Since $a_{ij} = \sum_{k=1}^{\min(i,j)} \ell_{ik} u_{kj}$, with $\ell_{ii} = 1$:

Row i of U ($i \le j$):

$$u_{ij} = a_{ij} - \sum_{k=1}^{i-1} \ell_{ik} u_{kj}$$

Column i of L ($j > i$, $u_{ii} \neq 0$):

$$\ell_{ji} = \frac{a_{ji} - \sum_{k=1}^{i-1} \ell_{jk} u_{ki}}{u_{ii}}$$

Stage i: compute row i of U, then column i of L (empty sums are zero).

Basic Algorithm and Existence Condition

Set $\ell_{ii} = 1$. For $i = 1, \ldots, n$:

1. compute u_{ij}, $j = i, \ldots, n$;
2. if $u_{ii} = 0$, stop (no pivot);
3. compute ℓ_{ji}, $j = i + 1, \ldots, n$.

Existence (nonsingular A)

All pivots are nonzero **iff** every leading principal minor is nonzero:

$$\Delta_k = \det A_{1:k,1:k} \neq 0, \quad k = 1, \ldots, n, \qquad u_{kk} = \Delta_k / \Delta_{k-1}.$$

Worked Example (No Pivoting)

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}.$$

Stage 1: $u_{11} = 2,\ u_{12} = 1,\ u_{13} = 1; \quad \ell_{21} = 2,\ \ell_{31} = -1.$

Stage 2: $u_{22} = -6 - 2 \cdot 1 = -8, \quad u_{23} = -2, \quad \ell_{32} = (-8)/(-8) = -1.$

Stage 3: $u_{33} = 2 - (-1)(1) - (-1)(-2) = 1.$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}, \qquad LU = A. \checkmark$$

Works here — but not for every nonsingular matrix.

Why Is It Called the Compact Form?

Known structure is free (L unit lower triangular, U upper triangular). Store both factors in **one** $n \times n$ array:

$$c_{ij} = \begin{cases} u_{ij}, & i \le j, \\ \ell_{ij}, & i > j, \end{cases} \qquad L = I + \operatorname{tril}(C, -1), \quad U = \operatorname{triu}(C).$$

$$C = \begin{bmatrix} 2 & 1 & 1 \\ 2 & -8 & -2 \\ -1 & -1 & 1 \end{bmatrix}$$

Storage	Scalars
Two full matrices	$2n^2$
Two triangular arrays	$n(n+1)$
Doolittle compact	$\mathbf{n^2}$

- Elimination link: multipliers sit where entries were zeroed, so the factorization can **overwrite** A in place.

Shortcoming 1: Zero Pivot despite Nonsingularity

$$A_0 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \det A_0 = -1 \neq 0.$$

First pivot $u_{11} = 0$, so $\ell_{21} = 1/0$ is undefined. If $A_0 = LU$ with unit lower L, the first column needs $u_{11} = 0$ *and* $\ell_{21} u_{11} = 1$ — impossible. Swapping rows gives

$$PA_0 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad L = I,\ U = PA_0.$$

Key point

A zero pivot does **not** imply singularity — just a bad row order.

Shortcoming 2: Small Pivot Amplifies Error

$$A_\varepsilon = \begin{bmatrix} \varepsilon & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad 0 < \varepsilon \ll 1.$$

Without swapping, $\ell_{21} = 1/\varepsilon$ is huge: $u_{22} = 1 - \frac{1}{\varepsilon}$, $y_2 = 2 - \frac{1}{\varepsilon}$.

With $\varepsilon = 10^{-5}$ and 4 significant digits both round to -1.000×10^5, so

$$\widehat{x}_2 = 1, \qquad \widehat{x}_1 = \frac{1 - \widehat{x}_2}{10^{-5}} = 0 \qquad (\text{exact } x_1, x_2 \approx 1).$$

Diagnosis

Not a divide-by-zero issue but *error propagation*; swapping makes the pivot 1.

The Fix: Partial Pivoting

At step k, pick the largest magnitude in the **current** column of $B^{(k)}$:

$$p \in \arg\max_{k \le i \le n} |b_{ik}^{(k)}|; \qquad \text{swap rows } k, p, \text{ then eliminate.}$$

- Uses the *updated* column, not the original column of A.
- A nonzero candidate gives a nonzero pivot; in exact arithmetic $|\ell_{ik}| \le 1$.

Limits

Bounding multipliers does not prevent element growth, nor cure ill-conditioning.

Partial-Pivoting Algorithm: $PA = LU$

P records row swaps; L unit lower triangular, U upper triangular. Left-multiplying by P only reorders *equations*.

Init $B \leftarrow A$, $L \leftarrow I_n$, $P \leftarrow I_n$. For $k = 1, \ldots, n$:

1. $p \in \arg\max_{k \le i \le n} |b_{ik}|$.
2. if $b_{pk} = 0$, stop (A singular).
3. if $p \neq k$: swap rows k, p of B and P; swap **only the first** $k-1$ **entries** of rows k, p of L.
4. for $i = k+1, \ldots, n$: $\ell_{ik} \leftarrow b_{ik}/b_{kk}$; $b_{ij} \leftarrow b_{ij} - \ell_{ik} b_{kj}$ $(j > k)$; $b_{ik} \leftarrow 0$.

Finally $U \leftarrow B$.

Why Swap the Already-Computed L?

- Columns $1..k-1$ of L record the multipliers each row used earlier; swapping rows of the working matrix swaps those histories.
- Mispaired multipliers **break** $PA = LU$.
- Swap **only** the finished first $k-1$ entries — swapping whole L rows would move the unit diagonal and break the structure.

Nonsingular A Always Has a Nonzero Pivot

Theorem

If A is nonsingular, in exact arithmetic every step of partial pivoting finds a nonzero pivot and terminates with $PA = LU$.

Proof.

Each step is a row swap (det ± 1) followed by unit lower-triangular elimination (det 1), so the working matrix keeps a nonzero determinant. Its leading part is upper triangular with the nonzero pivots on the diagonal, so in the block form $\begin{bmatrix} U_{11} & U_{12} \\ 0 & S_k \end{bmatrix}$ the trailing block satisfies $\det S_k \neq 0$. Hence S_k is nonsingular, its first column is not all zero, and the max-magnitude pivot is nonzero. Induction over k gives $PA = LU$. □

Conversely, an all-zero candidate column would give $\det S_k = 0 \Rightarrow \det A = 0$.

Example: Two Row Swaps

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}.$$

Step 1. Max in column 1 is 4: swap rows 1, 2; $\ell_{21} = \frac{1}{2},\ \ell_{31} = -\frac{1}{2}$:

$$B^{(2)} = \begin{bmatrix} 4 & -6 & 0 \\ 0 & 4 & 1 \\ 0 & 4 & 2 \end{bmatrix}, \quad L^{(2)} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ -\frac{1}{2} & 0 & 1 \end{bmatrix}.$$

Step 2. Column 2 ties at 4: choose row 3, swapping rows 2, 3 of B *and the finished first column of* L:

$$\widetilde{B} = \begin{bmatrix} 4 & -6 & 0 \\ 0 & 4 & 2 \\ 0 & 4 & 1 \end{bmatrix}, \quad \widetilde{L} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix} \Rightarrow\ \ell_{32} = 1,\ b_{33} = -1.$$

Result and Verification

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 1 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 4 & -6 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & -1 \end{bmatrix}, \quad P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad LU = \begin{bmatrix} 4 & -6 & 0 \\ -2 & 7 & 2 \\ 2 & 1 & 1 \end{bmatrix} = PA. \checkmark$$

Skipping the L swap is fatal

Swapping only B gives $L_{\text{wrong}} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ -\frac{1}{2} & 1 & 1 \end{bmatrix}$, for which $(L_{\text{wrong}}U)_{21} = 2 \neq -2 = (PA)_{21}$.

In-Place Compact Implementation

Let $C \leftarrow A$. At step k: columns $1..k-1$ hold L, rows $1..k-1$ hold U, and the trailing block holds S_k. Swapping rows k, p swaps **entire rows** of C (finished U rows are untouched since $k, p \geq k$).

$C \leftarrow A$, $\pi \leftarrow (1, \ldots, n)$. For $k = 1, \ldots, n$:

1. $p \in \arg\max_{k \leq i \leq n} |c_{ik}|$; if $c_{pk} = 0$, stop.
2. swap rows k, p of C and $\pi_k \leftrightarrow \pi_p$.
3. for $i > k$: $c_{ik} \leftarrow c_{ik}/c_{kk}$; $c_{ij} \leftarrow c_{ij} - c_{ik}c_{kj}$, $j > k$.

Read L, U back by the compact rule; $P_{i,\pi_i} = 1$, so $(PA)_{ij} = a_{\pi_i, j}$.

$$C = \begin{bmatrix} 4 & -6 & 0 \\ -\frac{1}{2} & 4 & 2 \\ \frac{1}{2} & 1 & -1 \end{bmatrix}, \qquad \pi = (2, 3, 1).$$

Solving $A\mathbf{x} = \mathbf{b}$ with $PA = LU$

$$A\mathbf{x} = \mathbf{b} \Rightarrow PA\mathbf{x} = P\mathbf{b} \Rightarrow LU\mathbf{x} = P\mathbf{b} \Rightarrow \boxed{L\mathbf{y} = P\mathbf{b}, \quad U\mathbf{x} = \mathbf{y}.}$$

With $\widetilde{b}_i = b_{\pi_i}$:

$$y_i = \widetilde{b}_i - \sum_{j=1}^{i-1} \ell_{ij} y_j, \qquad x_i = \frac{y_i - \sum_{j=i+1}^{n} u_{ij} x_j}{u_{ii}}.$$

Example

$\mathbf{b} = (4, -2, 7)^{\mathsf{T}} \Rightarrow P\mathbf{b} = (-2, 7, 4)^{\mathsf{T}}$: $y = (-2, 6, -1)^{\mathsf{T}} \Rightarrow x = (1, 1, 1)^{\mathsf{T}}$.

Cost and Stability

Work

- Factorization: $\approx \frac{2}{3}n^3$.
- Pivot search + swaps: $O(n^2)$; overall still $O(n^3)$.
- Each new $\mathbf{b}$: $O(n^2)$.

Stability

- Exact: no zero-pivot failure, $|\ell_{ik}| \leq 1$.
- Floating point: usually better, growth still possible.
- Ill-conditioned A stays sensitive.

Two different questions

Algebraic correctness $\neq$ floating-point stability. A tiny pivot justifies a warning, not a claim of singularity.

Conclusion

- The unit-diagonal convention gives recursive formulas and a **compact** n^2 storage that works in place.
- Zero or tiny pivots make the basic method fail or lose accuracy.
- **Partial pivoting** yields $PA = LU$, keeping the computed multipliers consistent on each row swap.
- Nonsingular A always completes in exact arithmetic, terminating with $PA = LU$.

References

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Thank you! Questions?